# ELEC5305 Project Proposal

## Speech Noise Reduction Using STFT and Wiener Filtering

## 1. Project Title

Speech Noise Reduction Using STFT and Wiener Filtering

## 2. Student Information

**Full Name:** Xinyu Shen

**Student ID (SID):** 530537427

**GitHub Username:** xinyu-shen-334

**GitHub Repository:** https://github.com/xinyu-shen-334/elec5305-project-530537427

**GitHub Project Site:** https://xinyu-shen-334.github.io/elec5305-project-530537427/

## 3. Project Overview

Background noise is a common problem in speech communication systems. Noise from traffic, fans, air conditioners, crowds, or other environmental sources can reduce the clarity and quality of recorded speech. This can affect phone calls, online meetings, voice recordings, hearing devices, and speech recognition systems.

The aim of this project is to develop a speech noise reduction system using the Short-Time Fourier Transform (STFT) and Wiener filtering. The noisy speech signal will first be divided into short time frames and transformed from the time domain into the time-frequency domain using the STFT. A Wiener filter will then be applied to reduce the estimated noise components while preserving important speech information. Finally, the enhanced speech signal will be reconstructed using the inverse STFT.

The effectiveness of the method will be evaluated by comparing the original noisy speech and the enhanced speech using waveform plots, spectrograms, and signal-to-noise ratio (SNR).

## 4. Background and Motivation

Speech enhancement is an important area of digital signal processing because real-world speech recordings usually contain unwanted background noise. A simple time-domain filter may not be effective when the noise and speech occupy similar frequency ranges. Therefore, time-frequency processing methods are commonly used to analyse and modify speech signals.

The Short-Time Fourier Transform is suitable for speech processing because speech is not completely stationary over a long period of time. The STFT divides a signal into short overlapping frames and calculates the frequency content of each frame. This makes it possible to observe how the frequency components of speech change over time.

Wiener filtering is a classical signal processing method that estimates a desired signal from a noisy observation by reducing the mean-square error. In a speech enhancement application, the filter can reduce frequency components that are likely to be dominated by noise while keeping components that are more likely to contain useful speech information.

This topic was selected because it combines several fundamental signal processing concepts, including Fourier analysis, time-frequency representations, filtering, noise estimation, and objective performance evaluation. It is also suitable for a semester project because it can be implemented and tested without requiring large neural networks or expensive computing resources.

## 5. Proposed Methodology

The project will be implemented using MATLAB or Python. A small collection of clean speech recordings will first be obtained from publicly available speech datasets or recorded directly for testing. Different types of background noise, such as white noise or environmental noise, will then be added to the clean recordings to generate controlled noisy speech signals.

- Load and normalise the speech signal.
- Add a selected background noise signal at a controlled noise level.
- Divide the noisy speech into overlapping short-time frames.
- Apply the STFT to obtain a time-frequency representation.
- Estimate the noise power spectrum from noise-dominated sections of the signal.
- Calculate and apply a Wiener filter in the frequency domain.
- Reconstruct the enhanced speech signal using the inverse STFT.
- Save and compare the original, noisy, and enhanced audio signals.

The system will be tested using different noise conditions and noise levels. The main quantitative evaluation metric will be SNR improvement. The waveform and spectrogram of each signal will also be plotted to show how noise components change before and after processing.

If time permits, different STFT window sizes or Wiener filter parameters will also be compared to investigate their effects on speech quality and noise reduction.

## 6. Expected Outcomes

The expected outcome is a working speech enhancement program that can take a noisy speech recording as input and generate an enhanced speech signal with reduced background noise.

- A working MATLAB or Python implementation.
- Example clean, noisy, and enhanced speech recordings.
- Waveform comparisons.
- Spectrogram comparisons.
- SNR measurements before and after processing.
- A discussion of the advantages and limitations of the Wiener filtering approach.
- Complete GitHub documentation and demonstration material.

A successful result should show a measurable increase in SNR and a visible reduction of noise components in the spectrogram while maintaining understandable speech.

## 7. Timeline

| Week | Task |
|---|---|
| Weeks 1–2 | Select project topic and define project objectives |
| Weeks 3–5 | Literature review and speech/noise dataset collection |
| Weeks 6–7 | Implement STFT analysis and signal visualisation |
| Weeks 8–9 | Implement Wiener filtering and inverse STFT reconstruction |
| Weeks 10–11 | Test different noise conditions and evaluate SNR performance |
| Week 12 | Analyse results and improve the implementation |
| Week 13 | Complete final report, GitHub documentation, and demonstration |

## 8. References


1. Boll, S. (1979). Suppression of acoustic noise in speech using spectral subtraction. IEEE Transactions on Acoustics, Speech, and Signal Processing, 27(2), 113–120.
2. Lim, J. S., & Oppenheim, A. V. (1979). Enhancement and bandwidth compression of noisy speech. Proceedings of the IEEE, 67(12), 1586–1604.
3. Ephraim, Y., & Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator. IEEE Transactions on Acoustics, Speech, and Signal Processing, 32(6), 1109–1121.


## 9. Appendix (Optional)

Additional figures, preliminary spectrograms, test signals, and experimental results may be added to this section as the project develops.